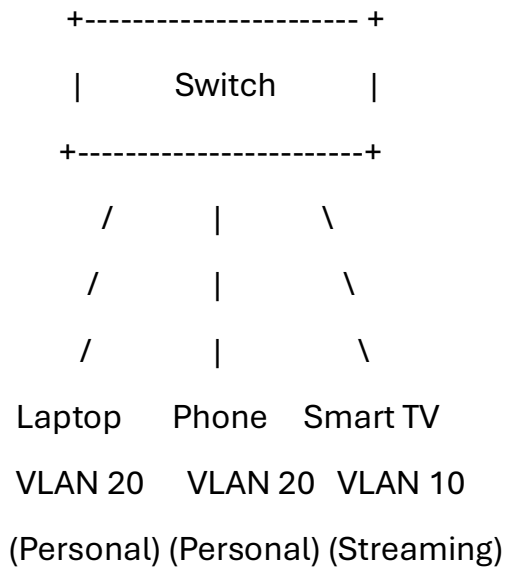


SWITCHING AND VLANS

1. Simple VLAN Network Diagram



VLAN Assignment

- VLAN 10 = Streaming devices (Smart TV)
- VLAN 20 = Personal devices (Laptop, Phone)

2. Create VLANs and Assign PCs

Switch Configuration

enable

configure terminal

vlan 10

name STREAMING

vlan 20

name PERSONAL

```
interface range fa0/1-3
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
interface range fa0/4-5
```

```
switchport mode access
```

```
switchport access vlan 20
```

```
end
```

PC Assignment

PC Port VLAN

PC1 Fa0/1 10

PC2 Fa0/2 10

PC3 Fa0/3 10

PC4 Fa0/4 20

PC5 Fa0/5 20

Verification

```
show vlan brief
```

Ping Test

- PC1 → PC2  Success
- PC1 → PC4  Fail
- PC2 → PC5  Fail

Different VLANs cannot communicate without routing.

3. Move a PC from VLAN 10 to VLAN 20

Before

interface fa0/3

switchport mode access

switchport access vlan 10

PC3 belongs to VLAN 10.

After

interface fa0/3

switchport mode access

switchport access vlan 20

PC3 now belongs to VLAN 20.

Connectivity Test

Before

- PC3 → PC1  Success
- PC3 → PC4  Fail

After

- PC3 → PC1  Fail
- PC3 → PC4  Success

4. Automate Assigning 10 Ports to VLAN 30

Configuration

enable

configure terminal

vlan 30

name VLAN30

```
interface range fa0/6-15
```

```
switchport mode access
```

```
switchport access vlan 30
```

```
end
```

Explanation

This uses the **interface range** command, so you do not need to configure all 10 ports individually.